

Anibal Estuardo Sierra Morales

asierra@astro.unam.mx | anibalsierram.github.io

Instituto de Astronomía - Universidad Nacional Autónoma de México - Ciudad de México

PROFESSIONAL EXPERIENCE

- **Associate Professor** Since November 2025
Instituto de Astronomía, Universidad Nacional Autónoma de México
Ciudad de México
- **Postdoctoral researcher in Astrophysics** June 2024 - October 2025
Mullard Space Science Laboratory, University College London
Working in the Planet Formation group of Dr. Paola Pinilla
London, England
- **Postdoctoral researcher in Astrophysics** April 2024 - May 2024
Pontificia Universidad Católica de Chile
Working in the group of Dr. Viviana Guzmán
Santiago de Chile, Chile
- **Postdoctoral Fondecyt Fellow** April 2022 - March 2024
Agencia Nacional de Investigación y Desarrollo (ANID)
Working in the group of Dr. Laura Pérez
Santiago de Chile, Chile
- **Postdoctoral researcher in Astrophysics** March 2020 - February 2022
Facultad de Ciencias Físicas y Matemáticas, Universidad de Chile
Working in the group of Dr. Laura Pérez
Santiago de Chile, Chile

EDUCATION

- **Ph.D. in Astrophysics** January 2016 - December 2019
Instituto de Radioastronomía y Astrofísica - Universidad Autónoma de México
Thesis: Dust dynamics and emission in protoplanetary disks. Advisor: Dr. Susana Lizano
Morelia, México
- **M.Sc. in Astrophysics** January 2014 - December 2015
Instituto de Radioastronomía y Astrofísica - Universidad Autónoma de México
Thesis: Dust dynamics in protoplanetary disks. Advisor: Dr. Susana Lizano
Morelia, México
- **B.Sc. in Physics** January 2008 - December 2013
Facultad de Ciencias Físicas y Matemáticas - Universidad de San Carlos de Guatemala
Thesis: Group representation in physics. Advisor: Ing. Rodolfo Samayoa
Guatemala, Guatemala

SELECTED HONORS AND AWARDS

- **Robert Boyd award** 2025
Space and Climate Physics, University College London
Logro científico destacado
United Kingdom
- **Ph.D. thesis award** 2021
División de Astrofísica de la Sociedad Mexicana de Física
Best Ph.D. thesis done in Mexico or by mexicans abroad (period 2020-21)
México
- **Alfonso Caso Medal** 2020
Universidad Autónoma de México
Best student of the generation, Ph.D. in Astrophysics
México
- **Alfonso Caso Medal** 2015
Universidad Autónoma de México
Best student of the generation, M.Sc. in Astrophysics
México
- **Cum Laude degree** 2013
Universidad de San Carlos de Guatemala
B.Sc. in Physics
Guatemala
- **University student excellence award** 2013
Universidad de San Carlos de Guatemala
Best student of the generation, Faculty of Engineering, Physics & Mathematics
Guatemala
- **Faculty student excellence award** 2009-2012
Universidad de San Carlos de Guatemala
Best physics student of the year for 4 years
Guatemala

SUPERVISED THESIS

- **co-Advisor M.Sc. Thesis in Astrophysics.** 2022-2023
Student: Luz Maritza Caballero Cano. Instituto de Radioastronomía y Astrofísica Morelia, México
Thesis: "Propiedades del polvo en el disco protoplanetario alrededor del sistema HD 142527"
- **Advisor M.Sc. Thesis in Physics.** 2022-to date
Student: Erick Alexander Rojas García. Universidad de San Carlos de Guatemala Guatemala, Guatemala
Thesis: "Impact of dust porosity on the radiative transfer properties in protoplanetary disks"

STUDENT SUPERVISION

- **Social Service Supervision (Universidad Nacional Autónoma de México)**
Students: Carlos Guillén (2026), Luis Gómez (2026)
- **Supervision of B.Sc. projects (Universidad Autónoma de Sinaloa)**
Student: Litzy Guzmán (2025)
- **co-Supervision of M.Sc. projects (Universidad Autónoma de México)**
Student: Luz Maritza Caballero (2022-2023)
- **Supervision of M.Sc. projects (Universidad de San Carlos de Guatemala)**
Student: Erick Rojas (2022-to date)
- **Supervision of B.Sc. projects (Universidad de Chile)**
Students: Benjamín Carrera (2023-2024), Benjamín Sotomayor (2022-2023)
- **Supervision of B.Sc. projects (Universidad de Chile)**
7 B.Sc. students between 2021-2024

TEACHING EXPERIENCE

- **Lecturer**
Graduate Program in Science (Astronomy) Ciudad de México, México
Radiative Processes in Astrophysics (Semester 2026-2)
- **Lecturer**
Preparatory courses, Graduate Program in Science (Astronomy) Ciudad de México, México
Classical Mechanics (Semester 2026-2)
- **Lecturer** 2020
Faculty of Physics and Mathematics, Universidad de Chile Santiago de Chile, Chile
General astronomy. Undergrad-level course for engineers and scientist students
- **Teaching assistant** 2019
Escuela Nacional de Estudios Superiores, Universidad Autónoma de México Morelia, México
Distributed computing. Undergrad-level course for science and technology students
- **Teaching assistant** 2018
Escuela Nacional de Estudios Superiores, Universidad Autónoma de México Morelia, México
Image analysis using Python. Undergrad-level course for science and technology students
- **Teaching assistant** 2013
Faculty of Engineering, Universidad Francisco Marroquín Guatemala, Guatemala
Calculus, linear algebra. Undergrad-level course for engineers students
- **Physics laboratory teacher** 2013
Physics Department, Universidad de San Carlos de Guatemala Guatemala, Guatemala
Classical Mechanics, Electromagnetism, and Optics. Undergrad-level laboratory for physics students

GRANTS AND SCHOLARSHIPS

- **FONDECYT fellow** 2022-2024
Agencia Nacional de Investigación y Desarrollo (ANID) Chile
- **CONACYT** 2016-2019
Consejo Nacional de Ciencia y Tecnología México
Scholarship for Ph.D. studies
- **CONACYT - OEA - AMEXCID** 2016
Organización de los Estados Americanos - Agencia Mexicana de Cooperación Internacional para el Desarrollo México
Ph.D. travel and accommodation

• CONACYT - OEI <i>Organización de los Estados Iberoamericanos</i> Ph.D. travel and accommodation	2016 México
• CONACYT <i>Consejo Nacional de Ciencia y Tecnología</i> Scholarship for M.Sc. studies	2014-2015 México
• CONACYT - OEA - AMEXCID <i>Organización de los Estados Americanos - Agencia Mexicana de Cooperación Internacional para el Desarrollo</i> M.Sc. travel and accommodation	2014 México
• USAC Grant <i>Universidad de San Carlos de Guatemala</i> Grant for B.Sc. studies	2008-2013 Guatemala

PROFESSIONAL SERVICE

• per-Reviewer <i>Scientific articles in ApJ, ApJL, and A&A. ~ 2 papers per year</i>	Since 2021
• Secretary <i>Central America and Caribbean association for Astronomy Sciences (Alpha-Cen)</i>	2020-2021
• Evaluation commission <i>Guatemalan Basic Sciences Olympiad</i>	2012-2013
• ad-Honorem teacher assistant <i>Faculty of Engineering, Universidad de San Carlos de Guatemala</i>	2011

OBSERVATIONAL EXPERIENCE

- **Awarded observation time:** ALMA and VLA (as PI and Co-I), APEX, and JWST (as Co-I)
Full list of observational time at anibalsierram.github.io/Research.html

PRESENTATIONS AND SCIENTIFIC TALKS

- **Colloquium and Seminar speaker** at over 12 different universities and observatories.
- **Contributed speaker** to over 6 national and international conferences
Full list of Seminar and contributed speaker at anibalsierram.github.io/Talks.html

PUBLIC OUTREACH

- **Speaker at multiple events for general public.** Guatemala, Mexico, and Chile. ~ 30 events.
- **Scientific press releases for ALMA Large program papers.** alma-maps.info/press.html, and AGE-PRO website in progress.

REFERENCES

1. **Susana Lizano**
Professor, Instituto de Radioastronomía y Astrofísica
Universidad Autónoma de México
Email: s.lizano@irya.unam.mx
M.Sc. and Ph.D. advisor
2. **Laura Pérez**
Professor, Departamento de Astronomía
Universidad de Chile
Email: lperez@das.uchile.cl
Postdoctoral supervisor
3. **Paola Pinilla**
Professor, Mullard Space Science Laboratory
University College London
Email: p.pinilla@ucl.ac.uk
Postdoctoral supervisor
4. **Myriam Benisty**
Director, Max Planck Institute for Astronomy
Email: benisty@mpia.de
Active Scientific collaborator